Advanced Optics Laboratory (PH4201)

Experiment: 03

Faraday Effect (Magneto-optic)

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March 21, 2026

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1 | Introduction

The Faraday effect, also known as Faraday rotation, is a magneto-optic phenomenon that demonstrates the fundamental interaction between light and a magnetic field. Discovered by Michael Faraday, this effect reveals that isotropic media become optically active when a magnetic field is applied parallel to the propagation direction of the light. This pivotal observation served as one of the earliest indications of the deep, underlying connection between optics and electromagnetism. Specifically, introducing a magnetic field in the direction of a light beam's propagation through a transparent medium induces a phenomenon known as circular birefringence.

As linearly polarized light travels through the optical medium under these conditions, its plane of polarization undergoes a measurable rotation. The angle of this rotation is a linear function of both the mean magnetic flux density and the length of the optical medium. The factor of proportionality governing this relationship is known as Verdet's constant. This specific experiment focuses on measuring and interpreting this magnetically induced circular birefringence.

2 | Objective

The objective of this experiment is to measure and interpret the magnetically induced circular birefringence known as the Faraday effect.

3 | Equipment Required

Here is the list of required equipment for the Faraday effect setup:

1. **Light Source:** Red diode lasers with wavelengths of 650 nm, secured on a laser mount.
2. **Polarizing Filters:** A polarizer rotator to polarize the incident light and a precision polarizer mount acting as an analyzer.
3. **Magnetic Field Generator:** An electromagnet formed by a long solenoid with 2508 turns of wire, connected to a magnet power supply. Number of turns per unit length is 167.2 /cm.
4. **Optical Medium:** A transparent flint glass rod, which serves as the sample placed inside the solenoid.
5. **Detection System:** A photodiode detector used to measure the intensity of the light beam after it passes through the analyzer.
6. **Mounting Apparatus:** A linear optical rail equipped with translating mounts. This track facilitates the secure attachment and precise spatial alignment of all optical components, ensuring the beam remains centered through the sample and upon the active area of the detector.

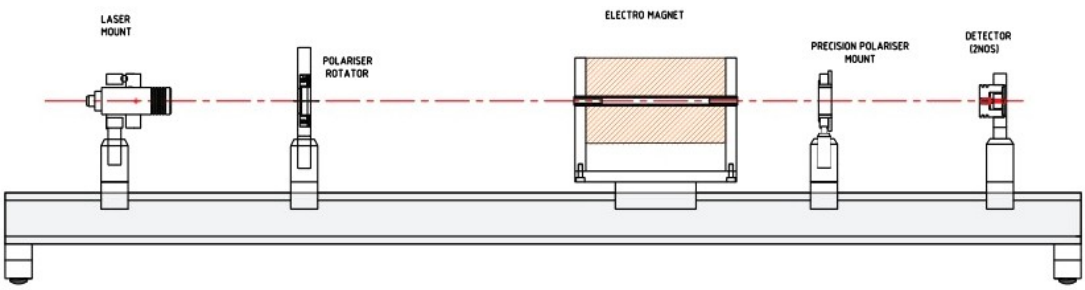

Figure 3.1: Schematic of Faraday setup

4 | Theory

4.1 | Optical Rotation for Linearly Polarized Light

A linearly polarized plane wave propagating along the x-direction can be expressed as a superposition of right and left circularly polarized components:

$$\mathbf{E}(x,t) = \mathbf{E}_r + \mathbf{E}_l \tag{4.1}$$

where

$$E_r(x,t) = \left\{0,\ \frac{A}{2}\cos\left[\omega\left(t-\frac{n_r x}{c}\right)\right],\ \frac{A}{2}\sin\left[\omega\left(t-\frac{n_r x}{c}\right)\right]\right\} \tag{4.2a}$$

$$E_l(x,t) = \left\{0,\ \frac{A}{2}\cos\left[\omega\left(t-\frac{n_l x}{c}\right)\right],\ -\frac{A}{2}\sin\left[\omega\left(t-\frac{n_l x}{c}\right)\right]\right\} \tag{4.2b}$$

represent the electric vectors in plane waves of right and left circularly polarized light, respectively. The amplitude of E is A, and the velocity of propagation is c/n where n is the index of refraction. Thus a beam of linearly polarized light can be considered as a linear superposition of right and left circularly polarized components.

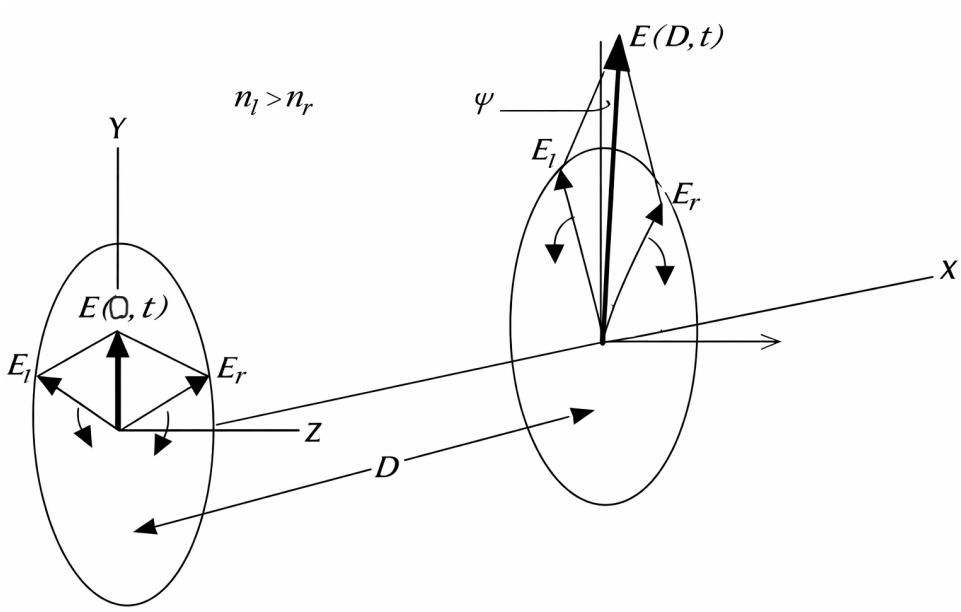

Figure 4.1: Addition of right and left circularly polarized components on two planes defined by $x = 0$ and $x = D$, respectively. Because $n_l > n_r$, E_l is more retarded in phase at $x = D$ relative to $x = 0$ than E_r, with the result that their sum at $x = D$ is rotated clockwise by the angle $\psi = \pi(n_l - n_r)D/\lambda$.

To indicate a difference in velocities for right and left circularly polarized waves the index n is replaced in equations (4.2a) and (4.2b) by n_r and n_l, respectively. As illustrated in Figure 4.1, the electric vector on the y-z plane at x=D is then

$$\boldsymbol{E}(D,t) = \boldsymbol{E}_r(D,t) + \boldsymbol{E}_l(D,t) \tag{4.3}$$

Using the identities

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \tag{4.4}$$

and

$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \tag{4.5}$$

we find for the electric field at $x = D$ the expression

$$\begin{aligned}\boldsymbol{E}(D,t) = \{0,\ & A\cos\left[\omega\left(t-(n_r+n_l)D/2c\right)\right]\cos\left[\omega(n_l-n_r)D/2c\right],\\ & A\cos\left[\omega\left(t-(n_r+n_l)D/2c\right)\right]\sin\left[\omega(n_l-n_r)D/2c\right]\}\end{aligned}$$

This represents an oscillating electric field of amplitude A inclined with respect to the y- axis by an angle

$$\psi = \tan^{-1}\left(\frac{E_z}{E_y}\right) = \frac{\omega(n_l-n_r)D}{2c} = \frac{\pi(n_l-n_r)D}{\lambda}, \tag{4.6}$$

where λ is the wavelength of the light in vacuum. Thus circular birefringence can be explained as an effect of a difference in the propagation velocities of right and left circularly polarized light. The difference is generally very

small, amounting to only a few parts per million in optically active materials as well as in the Faraday effect at moderate magnetic field strengths. However, in a typical setup $D/\lambda \gg 1$ so that even minute differences between n_l and n_r yield values of ψ which are large enough to be measured accurately.

$$\boxed{\psi = \frac{\pi D}{\lambda}(n_l - n_r)} \tag{4.7}$$

4.2 | Becquerel calculation on frequency shift

We consider a transparent medium composed of particles of mass m and charge q. These particles are embedded in a background of opposite charge and are bound by an elastic restoring force, allowing them to oscillate about fixed equilibrium positions.

We examine the case where a constant magnetic field $\mathbf{B}$ of magnitude B is applied along the $+x$-direction. A plane right circularly polarized electromagnetic wave propagates along the same direction. At a given point, the wave produces a time-varying electric field $\mathbf{E}$ of constant magnitude E, which rotates in a plane perpendicular to $\mathbf{B}$ and in a clockwise sense relative to the direction of $\mathbf{B}$.

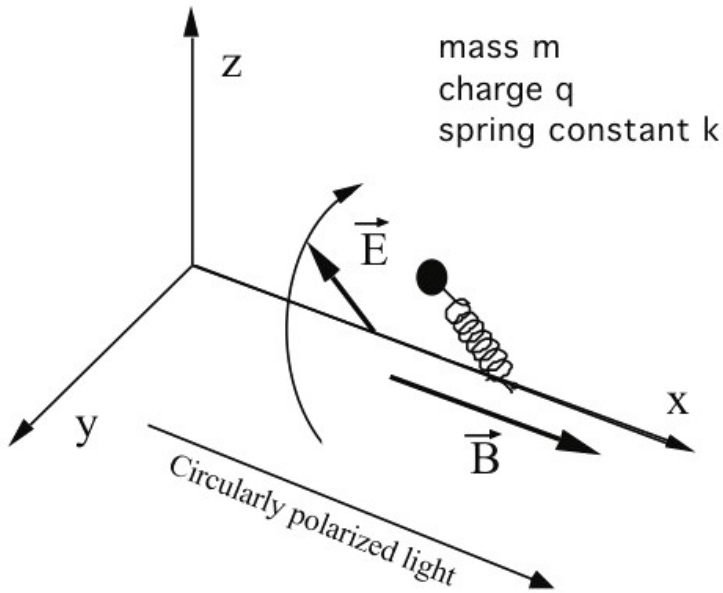

Figure 4.2: A particle of mass m and charge q is attached to a spring with spring constant k, which is fixed along the x-axis. The direction of propagation of light and the applied magnetic field $\mathbf{B}$ are both along the x-axis. The electric field $\mathbf{E}$ is perpendicular to the x-axis and rotates in the yz-plane.

Under *steady-state conditions*, the charged particles execute circular motion of radius r, governed by the equation (in cgs units):

$$-m\omega^2 r = -kr + qE + \frac{qB}{c}\omega r \tag{4.8}$$

where k represents the effective spring constant associated with the restoring force. For left circularly polarized light, the sign of the magnetic term changes.

Solving the above equation for r, we obtain

$$r = \frac{qE/m}{\omega_0^2 - \omega^2 - \frac{qB}{mc}\omega} \tag{4.9}$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}. \tag{4.10}$$

A displacement of a charge q from its oppositely charged equilibrium position by a distance r produces an electric dipole of magnitude qr. If there are N such dipoles per unit volume, the polarization is given by $P = Nqr$, and the permittivity becomes

$$\varepsilon = 1 + \frac{4\pi P}{E}.$$

Thus, for a dielectric medium with permeability $\mu = 1$, the velocity of propagation is $v = \frac{c}{\sqrt{\varepsilon}}$. In the presence of a magnetic field, the refractive index for right circularly polarized light is

$$n_r(\omega) = \frac{c}{v} = \left[1 + \frac{4\pi N q^2/m}{\omega_0^2 - \omega^2 - \frac{qB}{mc}\omega}\right]^{1/2} \tag{4.11}$$

and for left circularly polarized light,

$$n_l(\omega) = \frac{c}{v} = \left[1 + \frac{4\pi N q^2/m}{\omega_0^2 - \omega^2 + \frac{qB}{mc}\omega}\right]^{1/2}. \tag{4.12}$$

Above equations indicate that the refractive indices diverge as the frequency of light approaches the condition

$$\omega_0^2 - \omega^2 \pm \frac{qB}{mc}\omega = 0.$$

This unphysical behaviour arises from the absence of a damping term in the equation of motion. In a more realistic treatment, a damping force proportional to the particle velocity is included, which accounts for absorption. Both the refractive index and the absorption coefficient exhibit maxima near ω_0. However, for frequencies sufficiently far from ω_0, neglecting damping introduces negligible error.

The influence of the magnetic field on the refractive indices of circularly polarized light depends on the sign of the product qB. For right circularly polarized light with the magnetic field directed along the propagation direction ($B_x > 0$) and for negatively charged particles ($q < 0$), the product $qB < 0$. This leads to a decrease in n_r and a corresponding increase in n_l. Hence, $(n_l - n_r) > 0$, which implies a positive rotation angle.

The change in n_r due to the applied magnetic field can be interpreted as an effective shift in the frequency from ω to $\omega + \Delta\omega_r$, where $\Delta\omega_r$ satisfies

$$\omega^2 + \frac{qB}{mc}\omega = (\omega + \Delta\omega_r)^2. \tag{4.13}$$

Solving this equation and neglecting higher-order small terms gives

$$\boxed{\Delta\omega_r = \frac{qB}{2mc}.} \tag{4.14}$$

Similarly, for left circularly polarized light,

$$\omega^2 - \frac{qB}{mc}\omega = (\omega + \Delta\omega_l)^2, \tag{4.15}$$

which yields

$$\boxed{\Delta\omega_l = -\frac{qB}{2mc}.} \tag{4.16}$$

Since these frequency shifts are much smaller than ω, the difference in refractive indices can be expressed to a very good approximation using a *first-order expansion.*

$$n_l - n_r = \frac{dn}{d\omega}(\Delta\omega_l - \Delta\omega_r). \tag{4.17}$$

Substituting the expressions for $\Delta\omega_r$ and $\Delta\omega_l$, we obtain

$$n_l - n_r = \frac{dn}{d\lambda}\left(\frac{\lambda^2}{2\pi c}\right)\left(\frac{qB}{mc}\right), \tag{4.18}$$

where $\lambda = \frac{2\pi c}{\omega}$ is the wavelength of light in vacuum.

Combining Equation (4.6) and 4.18 the *Faraday rotation* angle is given by

$$\psi = \frac{dn}{d\lambda}\frac{\lambda}{2c^2}\frac{q}{m}DB. \tag{4.19}$$

Experimentally it was verified that the Faraday rotation angle ψ is proportional to the product of D and the magnetic field B, thus

$$\boxed{\psi = VDB,} \tag{4.20}$$

Thus, the **Verdet constant** can be written as by comparing

$$\boxed{V = \frac{dn}{d\lambda}\frac{\lambda}{2c^2}\frac{q}{m}.} \tag{4.21}$$

5 | Procedure

We have followed the following procedure to do this experiment:

1. First we placed the laser, P1 (polarizer), Faraday tube or that electro magnet on the optical rail (see Figure 5.1).
2. Then we did the alignment of the laser. To do so we first measured the height of the laser near the source and near the detector and matched it. We assured that the beam was completely passing through the tube (Electro magnet).
3. Then we changed the angle of the P1 to have maximum intensity in the detector. Note that we kept the current zero at this configuration and we did not place the P2 at the rail.
4. Next we put the P2 (polarizer) and made the configuration such that the intensity is minimum at the detector.
5. In this setup we did this experiment which had two part. In the first part we measured the variation of the angle of the rotation due to the magnetic effect and in the later part we measured the variation of the intensity over the current (i.e. magnetic field strength).

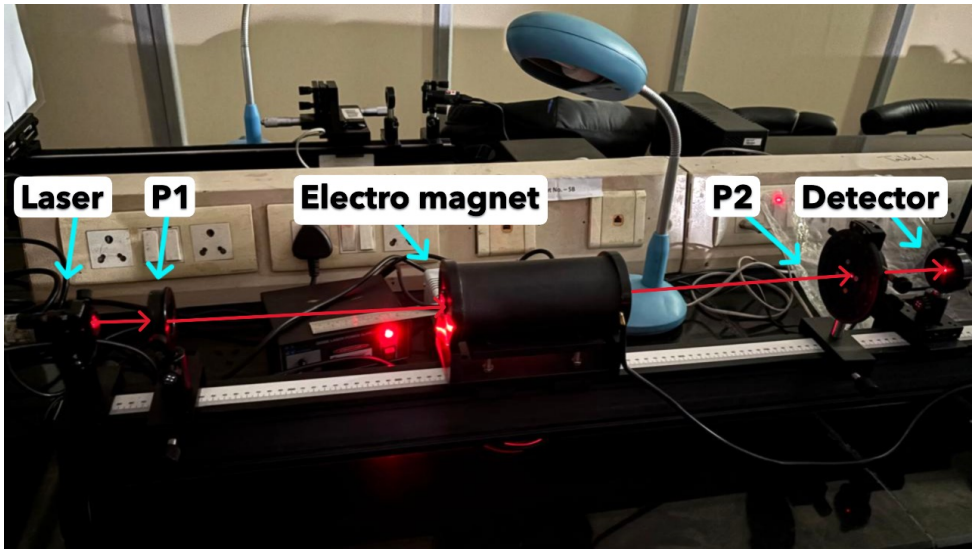

Figure 5.1: Setup for Faraday Effect

6 | Data analysis and calculation

As per the procedure we took the first reading using the setup (see Figure 5.1). In this setup we used the following parameters to calculated the magnetic field from the applied current.

- Number of turns in the coil: 2508.
- Number of turns per unit length: $N = \frac{2508}{15\text{ cm}} = 167.2\text{ cm}^{-1}$
- Refractive index of the glass rod: 1.717
- Wavelength of the laser: $\lambda = 650$ nm

And we used the following relation for the magnetic field

$$H = \pi N I \tag{6.1}$$

We measured the angle of rotation of the linearly polarized light using the P2. The method to measure the angle in the polarizer was similar to that of a Vernier caliper. Hence, the Least Count of polarizer $= 1^\circ - 0.9^\circ = 0.1^\circ$.

Table 6.1: Rotation of angle due to the Faraday-Magnetic effect

Current (A)	Magnetic field (Oe)	Theta in P2 (°)	Angle(Diff) (min)	Verdet constant
0	0	112	0	...
0.4	210.1097167	112.4	24	0.011422604
0.8	420.2194333	113.2	72	0.017133905
1.22	640.8346358	113.9	114	0.017789301
1.6	840.4388667	114.5	150	0.017847818
2.02	1061.054069	115.2	192	0.018095214
2.4	1260.6583	116	240	0.019037673
2.79	1465.515274	116.9	294	0.020061203
3.22	1691.383219	117.2	312	0.018446441
3.59	1885.734707	117.8	348	0.018454346
4.02	2111.602653	118.7	402	0.019037673
4.29	2253.426711	119.4	444	0.019703326

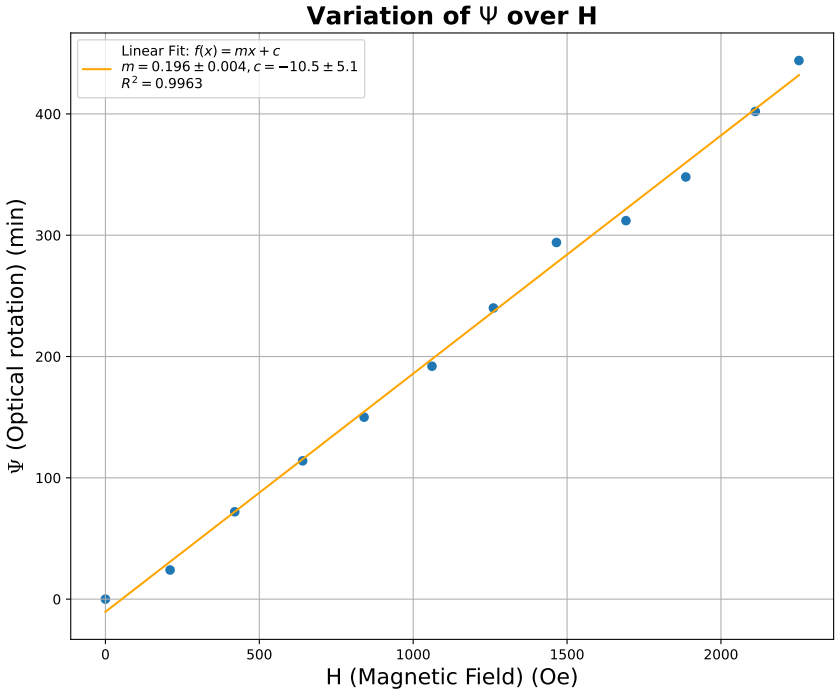

Figure 6.1: Variation of optical rotation angle over applied magnetic field

6.1 | Calculation of Verdet Constant

From the formula:

$$\Psi = VLH, \tag{6.2}$$

Here Ψ is optical rotation, V is the Verdet constant, L is the length of the glass, H is magnetic field. We know the Verdet constant V is given by:

$$V = \frac{\Psi}{L \cdot H} = \frac{\text{slope}}{\text{L}}. \tag{6.3}$$

From the slope of the Ψ vs H plot,

$$V = \frac{0.196 \pm 0.004}{10} = 0.0196 \pm 0.004 \text{ min Oe}^{-1} \text{ cm}^{-1},$$

For the second part, we tabulated for verifying Malus's Law. In this part we did not change the angle of the polarizer and only measure the intensity over the variation of the current.

Table 6.2: Table for variation of intensity over current variation

Current (A)	Magnetic field (Oe)	Intensity (μA)
0	0	0.8
0.4	210.1097167	1.3
0.84	441.230405	2.5
1.12	588.3072067	4.1
1.63	856.1970954	6.9
2.05	1076.812298	9.8
2.43	1276.416529	12.8
2.83	1486.526245	16.2
3.16	1659.866762	20.3
3.6	1890.98745	24.5
3.99	2095.844424	28.4

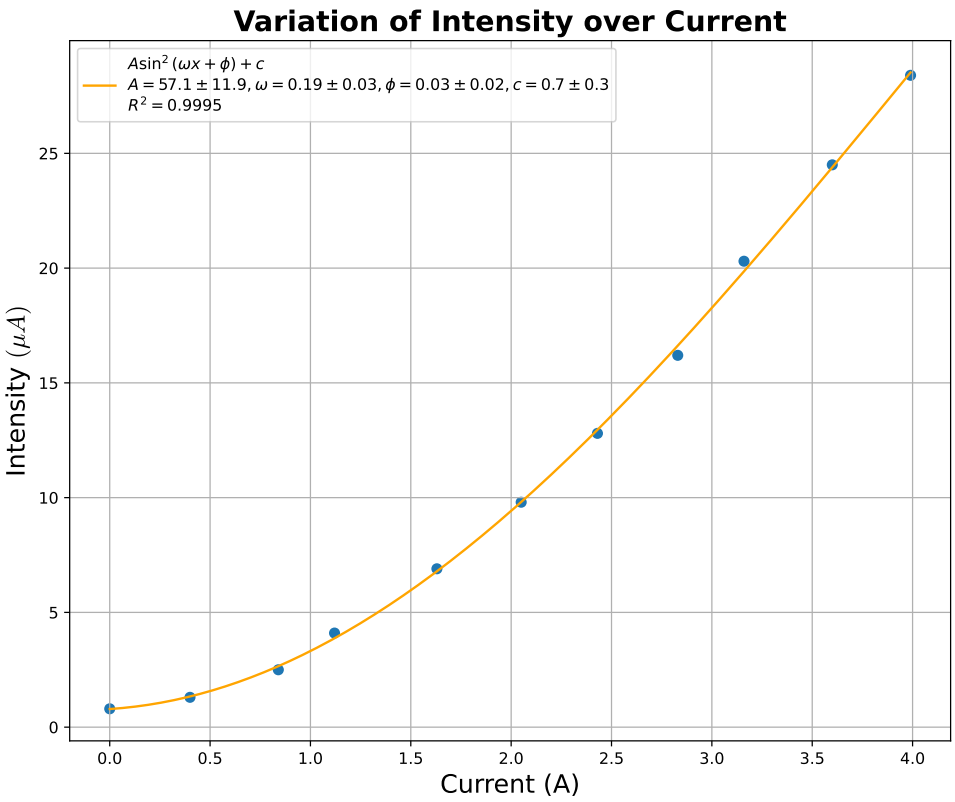

Figure 6.2: Variation of Intensity over current

6.2 | Verification for Malus's Law

According to Malus's law, the intensity of polarised light transmitted through an analyser is given by

$$I = I_0 \cos^2(\alpha), \tag{6.4}$$

where α represents the angle between the plane of polarisation and the transmission axis of the analyser.

In the presence of an applied magnetic field, the plane of polarisation undergoes a rotation of $\Psi = VHL$. Consequently, the theoretical expression for the transmitted intensity becomes

$$I_{\text{theoretical}} = I_0 \cos^2\left(\alpha_0 + VHL\right). \tag{6.5}$$

In the present setup, the P2 is initially oriented perpendicular to the plane of polarization. Due to this, an additional phase shift of $\alpha_0 = \frac{\pi}{2}$ is introduced, and the intensity follows a $\sin^2\theta$ dependence instead of $\cos^2\theta$. Also we write $H = \pi N i$. Hence the fitted curve is:

$$I(\text{Intensity}) = 57.1 \sin^2(0.19\, i(\text{current}) + 0.03) + 0.7$$

shows relation with the experimental data points, confirming the sinusoidal dependence of intensity. Thus, the experimental results successfully verify Malus's law within experimental uncertainty.

7 | Propagation of Error

The Verdet constant is given by

$$V = \frac{\theta}{HL} \tag{7.1a}$$

$$H = \pi N I \tag{7.1b}$$

where θ is the angle of rotation,H is magnetic field, I is the current, L is the length of the medium, and N is the number of turns per unit length.

Taking the natural logarithm on both sides,

$$\ln V = \ln\theta - \ln I - \ln L - \ln N \tag{7.2}$$

Differentiating, we obtain

$$\boxed{\frac{\Delta V}{V} = \frac{\Delta\theta}{\theta} + \frac{\Delta H}{H} + \frac{\Delta L}{L}} \tag{7.3}$$

we calculate the error for a data from the table of observations.

$$\begin{aligned} \theta &= 240 \text{ min}, \\ H &= 1260.6583 \text{ Oe}, \\ V &= 0.019037673 \text{ min Oe}^{-1}\text{cm}^{-1} \end{aligned}$$

The instrumental uncertainties are taken as:

$$\begin{aligned} \Delta\theta &= 0.1^\circ = 6 \text{ min}, \\ \Delta I &= 0.01 \text{ A}, \\ L &= 10 \text{ cm} \end{aligned}$$

Since $H = \pi N I$, the relative error in H is

$$\frac{\Delta H}{H} = \frac{\Delta I}{I} = \frac{0.01}{2.04} = 0.0049 \tag{7.4}$$

Now, substituting into the error formula (note that L is constant):

$$\frac{\Delta V}{V} = \frac{6}{240} + 0.0049 = 0.0299 \tag{7.5}$$

Hence, the maximum possible error is

$$\Delta V = 0.019037673 \times 0.0299 = 5.69 \times 10^{-4} \tag{7.6}$$

Therefore, the Verdet constant for this data point is $V = (1.90 \pm 0.057) \times 10^{-2}$ min Oe^{-1} cm^{-1}

8 | Sources of Error

8.1 | Thermal and Environmental Effects

- **Sample Heating:** The resistive heating of the electromagnet coil can transfer heat to the flint glass sample. This temperature increase can alter the Verdet constant of the glass and produce convection currents that locally vary the refractive index, introducing unwanted intensity changes.
- **Laser Intensity Drift:** Semiconductor diode lasers often exhibit slow temporal variations in output intensity, particularly before the laser reaches thermal equilibrium. If the laser is not allowed sufficient warm-up time, this drift can be mischaracterized as a change in transmitted intensity due to polarization rotation.

8.2 | Instrumental and Alignment Limitations

- **Initial Alignment and Zero Error:** The most significant systematic error arises from the initial optical alignment. If the polarizers P1 and P2 are not perfectly crossed when the magnetic field is at zero, a baseline offset is introduced. This accounts for the non-zero intercept observed in the linear regression.
- **Detector Non-linearity and Spatial Sensitivity:** The photodiode signal may not respond perfectly linearly to incident light intensity. Furthermore, if the laser beam is not strictly centered on the active area of the photodiode, any slight beam deflection caused by rotating the optical components can cause spurious intensity readings.
- **Magnetic Field Non-uniformity:** The standard theoretical formula $\varphi = VHL$ assumes a perfectly uniform magnetic field H across the entire length of the optical medium. In reality, the field lines begin to diverge near the ends of the solenoid, meaning the magnetic field is not perfectly constant across the entire length of the glass rod.

8.3 | Observational Uncertainty

- **Parallax Error in Angular Measurement:** The physical design of the setup, specifically the proximity of the electromagnet's cylindrical housing to the analyzing polarizer, restricts the observer's direct line of sight to the graduated scale. This obstruction forces the experimenter to read the angle from an oblique viewing position, introducing a parallax error that reduces the precision of the rotational measurements.
- **Resolution of the Null Point:** Accurately determining the exact angle of minimum transmitted intensity (the "darkness" point) is limited by the mechanical resolution (least count) of the precision polarizer mount. Additionally, at near-zero intensity, the signal-to-noise ratio of the photodiode drops, making the exact minimum difficult to pinpoint precisely.
- **Backlash Error:** Mechanical play or backlash in the rotational mount of the analyzing polarizer can introduce discrepancies if the minimum intensity angle is approached from different directions (clockwise vs. counter-clockwise). Though while taking the reading we maintain single convention (clockwise) to make the P2 for minimum intensity.

9 | Discussion

To overcome the limitations identified in the preceding section and to improve experimental precision, several refinements can be proposed. Replacing the manual analyser with a high-precision motorised rotation stage integrated with a lock-in amplifier would minimise human reading errors, suppress ambient light interference, and significantly enhance angular resolution.

Furthermore, rather than depending on a pre-calibrated curve for the electromagnet, the implementation of an *in situ* Hall probe (Gaussmeter) would enable real-time and accurate measurement of the magnetic field directly at the sample location.

Thermal effects can be mitigated by incorporating a water-cooling jacket or a thermoelectric (Peltier) cooling system around the sample holder. Such measures would ensure proper thermal isolation from the electromagnet and maintain a stable temperature, thereby allowing the Verdet constant to be determined under controlled conditions.

The present experimental arrangement also provides a robust platform for extended investigations. According to Becquerel's formulation, the Verdet constant exhibits an intrinsic dependence on wavelength, given by

$$V \propto \frac{dn}{d\lambda}. \tag{9.1}$$

By replacing the single diode laser with a tunable laser source or a white light source combined with narrow bandpass filters, one can experimentally determine the Verdet constant across multiple wavelengths and thereby study the dispersion characteristics of the material.

In addition, the flexibility of the optical rail facilitates the substitution of different active media. Investigating materials such as diamagnetic, paramagnetic, or even liquid samples (e.g., distilled water or saline solutions) would enable a comparative analysis of how variations in electronic structure and spin configurations influence both the magnitude and the direction (sign) of Faraday rotation.

10 | Conclusion

In this experiment, the Faraday magneto-optic effect was successfully observed and quantitatively analysed. The results demonstrated a clear linear dependence between the applied longitudinal magnetic field and the rotation angle of the plane of polarisation. This behaviour was confirmed through a precise linear fit, yielding a coefficient of determination of $R^2 \approx 0.996$, indicating excellent agreement with theoretical expectations.

From this linear relationship, the Verdet constant of the medium was experimentally determined as

$$V = (1.96 \pm 0.004) \times 10^{-2} \ \text{min Oe}^{-1}\ \text{cm}^{-1}. \tag{10.1}$$

In addition, the variation of transmitted light intensity with increasing current (magnetic field strength) was systematically studied. The observed intensity profile closely followed the theoretical $\sin^2$ dependence, thereby providing strong experimental validation that the magnetically induced rotation of polarised light is consistent with Malus's law.